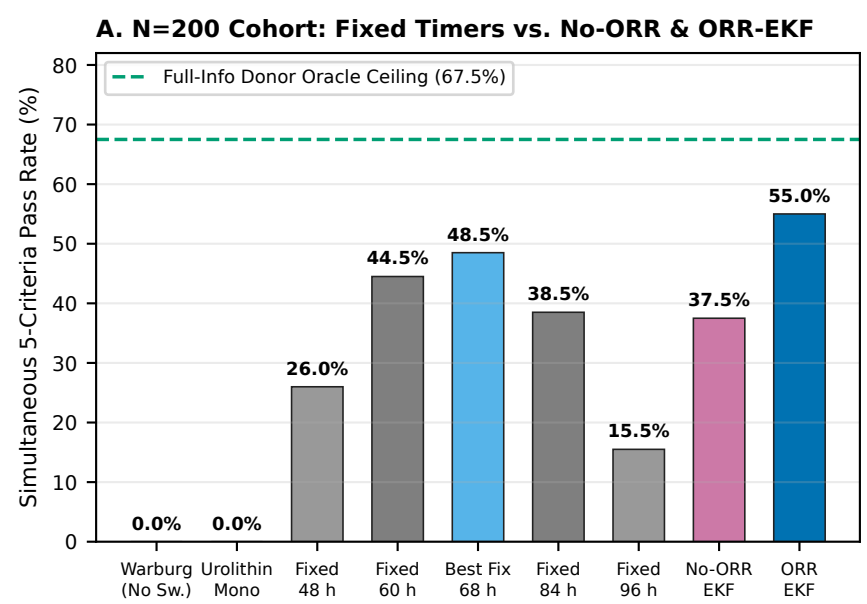
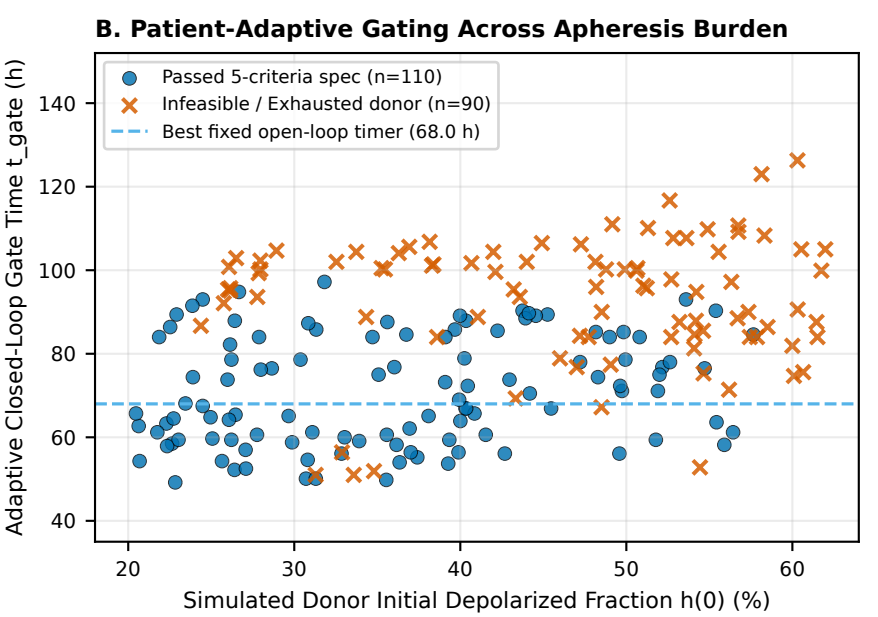
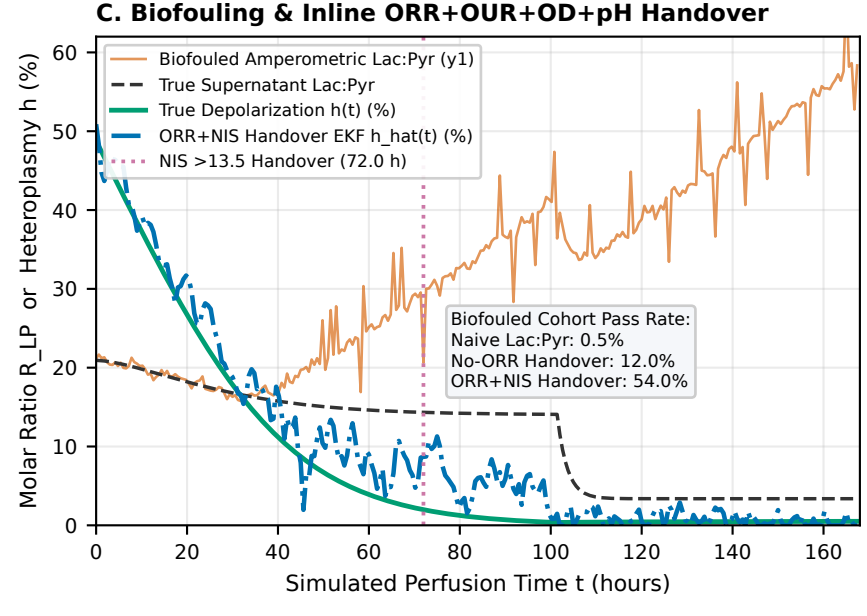
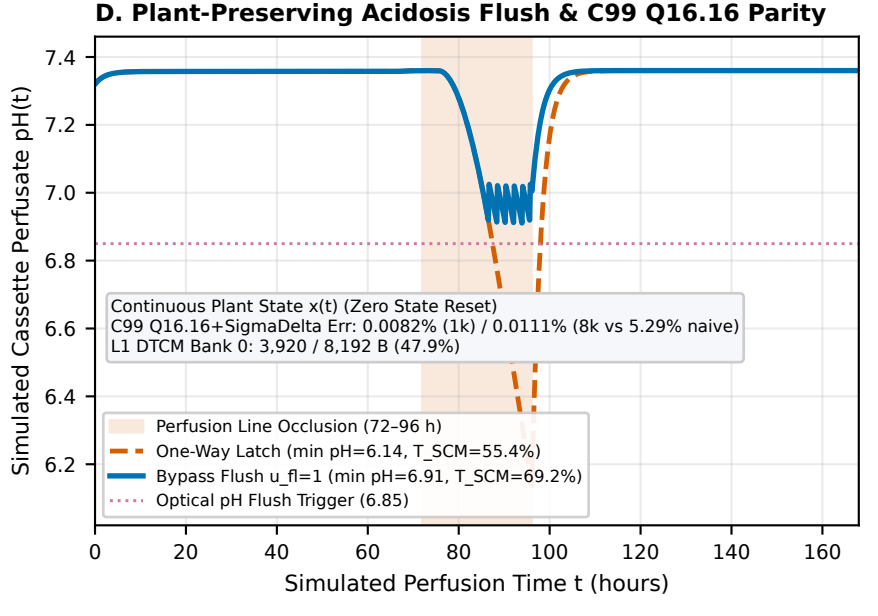

A. N=200 Cohort: Fixed Timers vs. No-ORR & ORR-EKF
Simultaneous 5-Criteria Pass Rate (%)
Full-Info Donor Oracle Ceiling (67.5%)
0.0%
0.0%
26.0%
44.5%
48.5%
38.5%
15.5%
37.5%
55.0%
Warburg (No Sw.)
Urolithin Mono
Fixed 48 h
Fixed 60 h
Best Fix 68 h
Fixed 84 h
Fixed 96 h
No-ORR EKF
ORR EKF


B. Patient-Adaptive Gating Across Apheresis Burden
Passed 5-criteria spec (n=110)
Infeasible / Exhausted donor (n=90)
Best fixed open-loop timer (68.0 h)
Adaptive Closed-Loop Gate Time t_gate (h)
Simulated Donor Initial Depolarized Fraction h(0) (%)


C. Biofouling & Inline ORR+OUR+OD+pH Handover
Biofouled Amperometric Lac:Pyr (y1)
True Supernatant Lac:Pyr
True Depolarization h(t) (%)
ORR+NIS Handover EKF h_hat(t) (%)
NIS >13.5 Handover (72.0 h)
Biofouled Cohort Pass Rate:
Naive Lac:Pyr: 0.5%
No-ORR Handover: 12.0%
ORR+NIS Handover: 54.0%
Molar Ratio R_LP or Heteroplasmy h (%)
Simulated Perfusion Time t (hours)


D. Plant-Preserving Acidosis Flush & C99 Q16.16 Parity
Continuous Plant State x(t) (Zero State Reset)
C99 Q16.16+SigmaDelta Err: 0.0082% (1k) / 0.0111% (8k vs 5.29% naive)
L1 DTCM Bank 0: 3,920 / 8,192 B (47.9%)
Perfusion Line Occlusion (72–96 h)
One-Way Latch (min pH=6.14, T_SCM=55.4%)
Bypass Flush u_fl=1 (min pH=6.91, T_SCM=69.2%)
Optical pH Flush Trigger (6.85)
Simulated Cassette Perfusate pH(t)
Simulated Perfusion Time t (hours)